

Technical Report

W. Chan

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Supervisors:	Dr. E.J. van Kampen
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E-mail:	w.y.chan@student.tudelft.nl

IDHP Step by Step

Table 1: IDHP step by step, α tracking task for short period model, target critic used.

Variable	t_0	t_1	t
Env reward, c	$c_0 =$ $-\frac{1}{2}(\alpha_{0,ref} - \alpha_0)^2$	$c_1 =$ $-\frac{1}{2}(\alpha_{1,ref} - \alpha_1)^2$	$c_t =$ $-\frac{1}{2}(\alpha_{t,ref} - \alpha_t)^2$ (1.1)
Reward grad, $\frac{\partial c}{\partial x}$	$\frac{\partial c_0}{\partial x_0} =$ $-(\alpha_{0,ref} - \alpha_0)$	$\frac{\partial c_1}{\partial x_1} =$ $-(\alpha_{1,ref} - \alpha_1)$	$\frac{\partial c_t}{\partial x_t} =$ $-(\alpha_{t,ref} - \alpha_t)$ (1.2)
Env obs, s	$s_0 =$ $(\alpha_{0,ref} - \alpha_0)$	$s_1 =$ $(\alpha_{1,ref} - \alpha_1)$	$s_t =$ $(\alpha_{t,ref} - \alpha_t)$ (1.3)
Model state, x	$x_0 =$ $[\alpha_0 \quad q_0]^\top$	$x_1 =$ $[\alpha_1 \quad q_1]^\top$	$x_t =$ $[\alpha_t \quad q_t]^\top$ (1.4)
Env/Model action, a	$a_0 =$ $[\delta_{0,e}]$	$a_1 =$ $[\delta_{1,e}]$	$a_t =$ $[\delta_{t,e}]$ (1.5)
RL			
Critic/TD error, e	$e_0 =$ <i>None</i>	$e_1 =$ $\lambda(x_0) - \frac{\partial c_0}{\partial x_0} - \gamma \lambda'(x_1) \frac{\partial x_1}{\partial x_0}$	$e_t =$ $\lambda(x_{t-1}) - \frac{\partial c_{t-1}}{\partial x_{t-1}} - \gamma \lambda'(x_t) \frac{\partial x_t}{\partial x_{t-1}}$ ¹ (1.6)
Critic weight, w_c	$w_{0,c} =$ <i>init</i>	$w_{1,c} =$ $w_{0,c} - \eta_c e_0^\top \frac{\partial \lambda(x_0)}{\partial w_{0,c}}$	$w_{t,c} =$ $w_{t-1,c} - \eta_c e_{t-1}^\top \frac{\partial \lambda(x_{t-1})}{\partial w_{t-1,c}}$ (1.7)
Target critic, $w_{c'}$	$w_{0,c'} =$ $w_{0,c}$	$w_{1,c'} =$ $\tau w_{1,c} + (1 - \tau) w_{0,c'}$	$w_{t,c'} =$ $\tau w_{t,c} + (1 - \tau) w_{t,c'}$ (1.8)
Actor loss, L	$L_0 =$ $c_0 + \gamma J(x_0)$	$L_1 =$ $c_1 + \gamma J(x_1)$	$L_t =$ $c_t + \gamma J(x_t)$ (1.9)
Actor weight, w_a	$w_{0,a} =$ <i>init</i>	$w_{1,a} =$ $w_{0,a} - \eta_a (\frac{\partial c_0}{\partial x_1} + \gamma \lambda(x_1)) \frac{\partial x_1}{\partial a_0} \frac{\partial a_0}{\partial w_{0,a}}$	$w_{t,a} =$ $w_{t-1,a} - \eta_a (\frac{\partial c_{t-1}}{\partial x_t} + \gamma \lambda(x_t)) \frac{\partial x_t}{\partial a_{t-1}} \frac{\partial a_{t-1}}{\partial w_{t-1,a}}$ (1.10)
RLS			
δx	$\delta x_0 =$ <i>None</i>	$\delta x_1 =$ $x_1 - x_0$	$\delta x_t =$ $x_t - x_{t-1}$ (1.11)
δa	$\delta a_0 =$ <i>None</i>	$\delta a_1 =$ $a_1 - a_0$	$\delta a_t =$ $a_t - a_{t-1}$ (1.12)

¹Using Zhou's TD convention, reward denoted as the earlier timestep.

X	$X_0 =$ $None$	$X_1 =$ $[\partial x_1 \quad \partial a_1]^\top$	$X_t =$ $[\partial x_t \quad \partial a_t]^\top$	(1.13)
RLS gain, k	$k_0 =$ $None$	$k_1 =$ $\frac{\Sigma_0 X_1}{\gamma_R + X_1^\top \Sigma_0 X_1}$	$k_t =$ $\frac{\Sigma_{t-1} X_t}{\gamma_R + X_t^\top \Sigma_{t-1} X_t}$	(1.14)
Estimated $\hat{\delta}x$	$\hat{\delta}x_0 =$ $None$	$\hat{\delta}x_1 =$ $X_1^\top \Theta_0$	$\hat{\delta}x_t =$ $X_t^\top \Theta_{t-1}$	(1.15)
RLS error, ϵ	$\epsilon_0 =$ $None$	$\epsilon_1 =$ $\delta x_1 - \hat{\delta}x_1$	$\epsilon_t =$ $\delta x_t - \hat{\delta}x_t$	(1.16)
Estimated param, Θ	$\Theta_0 =$ $init$	$\Theta_1 =$ $\Theta_0 + k_1 \epsilon_1$	$\Theta_t =$ $\Theta_{t-1} + k_t \epsilon_t$	(1.17)
RLS cov, Σ	$\Sigma_0 =$ $init$	$\Sigma_1 =$ $\frac{1}{\gamma_R}(\Sigma_0 - k_1 X_1^\top \Sigma_0)$	$\Sigma_t =$ $\frac{1}{\gamma_R}(\Sigma_{t-1} - k_t X_t^\top \Sigma_{t-1})$	(1.18)